

3. Set of measurement units which are controlled by the test controller and reads output from the embedded system under test.

4. While generation of signals addresses the input and output of products, user interfaces like keyboards, knobs, levers, and displays such as LCD and LED need special visual equipment like camera with image processing capability, solenoid, and robotic units for operating the switches, knobs, etc.

While these four modules are pretty much common to any embedded system testing, how the testers are built, configured, and used depends on the manufacturing phase of the embedded system. Let us briefly see the different testers that are used in different phases of the manufacturing process.

Electronics PCBA testing: in-circuit testing

Referring to Fig. 3, we can see that there are four stages of testing needed before the product is shipped out. In the first stage, the PCBA (raw PCB loaded with the components and soldered) is tested independently. This test is done when the board comes out of the soldering stage in the manufacturing line (after the inspection for any visible defects).

However, just PCBAs alone do not make the full embedded system as there are many more components needed. This essentially means that there are further processes after the

PCBAs are tested.

The PCBAs have to be tested individually before these are assembled into the full system. Due to this, PCBAs are tested using in-circuit tester (ICT).

In PCBA stage, only limited functionalities of the overall system can be tested. Additionally, PCBAs may have connectivity issues due to imperfections in the soldering process.

In-circuit testers focus on testing the components and interconnects in the PCBAs. This ensures all the components soldered in the board are working and also that the traces between the components are soldered well and working before it is integrated into the system.

We need to remember that ICT-tested PCBA does not guarantee that the final product will work when assembled fully as there are other components like cables, displays, keys, and power supplies which also need to be tested and faultless. ICT testing ensures that the PCBAs are tested to be free of defects before these are assembled into the system.

In-circuit testers operate on the principle of back-driving (apply stimulus) and measure the response from the node that is tested. Fig. 6 shows a simplified view of the in-circuit tester. These are similar to generic tester in architecture. They have a set of pins to drive the signals and a measuring circuit connected to the same pins to read the response.

A unique feature of the ICT is that the PCBAs are tested using pins in the test fixture, which directly probe the

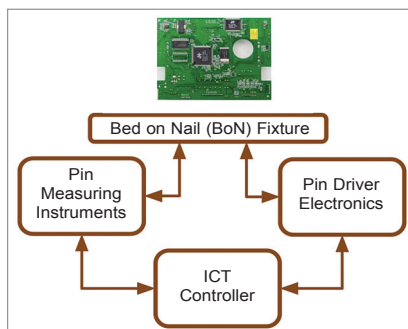


Fig. 6: Simplified view of the in-circuit tester

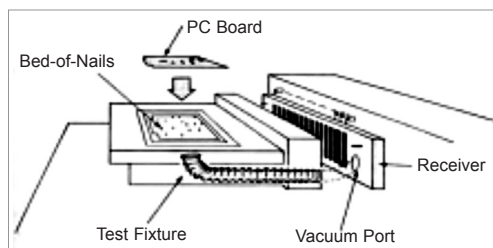


Fig. 8: Concept of ICT test fixture and a typical ICT tester

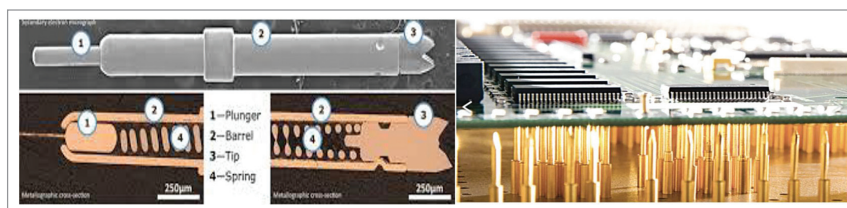


Fig. 7: Cross-sections of a typical pogo pin and how the bed-of-nail jig is implemented

test pads provided in the PCB for testing. For every unique PCBA, a fixture has to be developed to access the test pads provided through spring-loaded pins called pogo pins (named after the pogo sticks that children use for jumping and playing).

Fig. 7 shows a typical pogo pin cross-section and the cross-section of how the bed-of-nail jig is implemented for better understanding. Structure of the test jig has three components—a base plate in which the pogo pins are located at the locations where the PCBAs under test have test pads, a connection termination board where the pins are connected to the ICT drive and measurement electronics, and a vacuum cover which presses the PCBAs down on to the pogo pins during the test process. Fig. 8 shows the concept of ICT test fixture and a typical ICT tester for better understanding.

Since the ICTs use a plate with pogo pins for testing, they are also called bed-of-nail (BoN) testers. These pins are colloquially called nails!

Tester vendors provide test libraries and ICT test program developers use these to write the test program to test the PCBAs. The biggest advantage of ICTs is that they eliminate nearly hundred percent of the faulty PCBAs before they go to the next stage in the manufacturing line.

ICTs operate on the principle of comparing the signatures captured from a set of good tested boards called gold boards and compare the results with the boards under test.

These ICT testers are expensive. A new BoN jig has to be developed with corresponding test program for testing every new version of the PCBA, which increases the manufacturing cost.

One strategy is that for low-volume products, the ICT phase is skipped and designers use functional testers to test the product post assembly. While the risk of a faulty PCB leading to test-failure is high, the costs can thus be kept down due to low volume.

Sub-system and assembled product testing – ATE

The second and third tests cover the sub-systems and systems. These tests are executed by specially built custom